

Unsupervised Learning Project Report

Retail Consultant Analysis

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Abstract

This report presents an end-to-end unsupervised machine learning pipeline applied to the Online Retail II dataset. The pipeline spans feature engineering, customer segmentation via PCA and clustering, segment-specific market basket analysis, time-series decomposition, anomaly detection using Local Outlier Factor, and a collaborative-filtering recommendation system based on Singular Value Decomposition. The analysis identifies two distinct customer segments, uncovers high-confidence cross-sell rules, quantifies weekly demand seasonality, and flags 59 anomalous customers for business review.

Prologue: The Business Problem

This project was conducted from the perspective of a data science consultant hired by a medium-sized online retailer. The company possesses extensive transactional data but lacks structured insight into its customers and purchasing behaviour. Specifically, the retailer cannot identify its most valuable customers, does not know which products are commonly purchased together by different customer types, and is concerned about unusual or potentially fraudulent purchasing patterns.

The objective of this project is to construct an end-to-end unsupervised learning pipeline that transforms raw transactional data into actionable business insights. The pipeline proceeds sequentially: raw transactions are cleaned and aggregated into customer-level features; customers are segmented using dimensionality reduction and clustering; each segment is analysed for product associations; the top-selling item's demand is decomposed into trend, seasonal, and residual components; anomalous customers are identified using density-based outlier detection; and finally a latent-factor recommendation model is trained and evaluated.

1 Data Engineering and Feature Extraction

1.1 Preprocessing

The raw dataset (Online Retail II, sourced from the UCI Machine Learning Repository) contains one row per item purchased per invoice. After loading, the following cleaning steps were applied. Rows with missing **Customer ID** were dropped, as these transactions cannot be attributed to any customer. Invoices beginning with **C** (credit notes and returns) were excluded, since they represent reversals rather than genuine purchases. Rows with **Quantity** ≤ 0 or **Price** ≤ 0 were removed to eliminate data entry errors and free samples. After cleaning, the dataset contained 805,549 rows across 9 columns. A **Revenue** column was computed as **Quantity** $\times$ **Price**, and **InvoiceDate** was parsed with **dayfirst=True**, consistent with the European date formatting of this dataset.

1.2 Customer-Centric Feature Table

The cleaned transactional data was aggregated to produce one row per **Customer ID**, yielding a table of **5,878 customers** described by six features. The median values across all customers are summarised in Table 1.

Table 1: Median Customer Features

Feature	Median Value
Recency	96.00
Frequency	3.00
Monetary	898.92
Variety	45.00
AvgOrderValue	285.07
AvgQuantityPerInvoice	157.71

These statistics show that the typical customer purchases infrequently (3 invoices) but spends a moderate amount (£899 total) and buys a diverse set of 45 product types. The median **AvgQuantityPerInvoice** of 157.71 units hints that some customers are ordering in bulk even in the middle of the distribution, foreshadowing the wholesale segment identified in Section 2.

The four mandatory features are defined as follows. **Recency** is the number of days since the customer’s last purchase, computed as the difference between a reference date (maximum invoice date plus one day) and the customer’s most recent invoice; a lower value indicates more recent activity. **Frequency** is the number of unique invoices placed by the customer, capturing how often they buy independently of how much. **Monetary** is the total revenue generated by the customer across all purchases and is the primary measure of customer value. **Variety** is the number of unique stock codes purchased, capturing the breadth of purchasing behaviour: a customer who buys across many product categories is behaviourally very different from one who purchases the same few items repeatedly, even at similar total spend.

The two self-chosen features are defined as follows. **AvgOrderValue** is Monetary divided by Frequency. This feature distinguishes high-spending-per-visit customers from frequent but low-value customers. Two customers with identical Monetary values can have very different purchasing personalities depending on how spending is distributed across visits. **AvgQuantityPerInvoice** is total quantity purchased divided by Frequency. This captures bulk-buying behaviour per transaction: a wholesale buyer placing orders of 500 units per invoice is structurally different from a retail buyer ordering 3–4 items, making this feature especially valuable for anomaly detection.

2 Customer Segmentation and Visualization

2.1 Preprocessing for Clustering

Since all clustering algorithms used here are distance-based, proper preprocessing is critical. Two steps were applied. First, a log transformation (`np.log1p`) was applied because customer spending and quantity data are highly right-skewed: a small number of wholesale buyers generate orders of magnitude more revenue than the average customer, and

log transformation compresses this skew to prevent extreme values from dominating distance calculations. Second, `StandardScaler` was applied to standardise all features to zero mean and unit variance, ensuring that no single feature dominates the Euclidean distance metric.

2.2 PCA Projection

PCA was applied to reduce the six-dimensional scaled feature space to two components for visualisation. PC1 explains 64.7% of variance and PC2 explains 21.4%, together capturing 86.1% of total variance. This is a strong reduction, indicating that the customer data is largely structured along two dominant axes of variation.

The PCA scatter plot (Figure 1) reveals a fan-shaped distribution: the bulk of customers cluster tightly near the origin, while a smaller set stretches toward higher PC1 values, already suggesting the two-group structure that clustering will confirm. The PCA loadings show that PC1 is dominated by `AvgQuantityPerInvoice` and `Monetary`, meaning it separates customers primarily by order volume and total spend. PC2 is driven more by `Recency` and `Frequency`. Customers at the extreme high end of PC1 are therefore those who place enormous individual orders — exactly the Wholesale Buyers the pipeline identifies. Distance along PC1 is essentially a measure of wholesale-versus-retail purchasing behaviour.

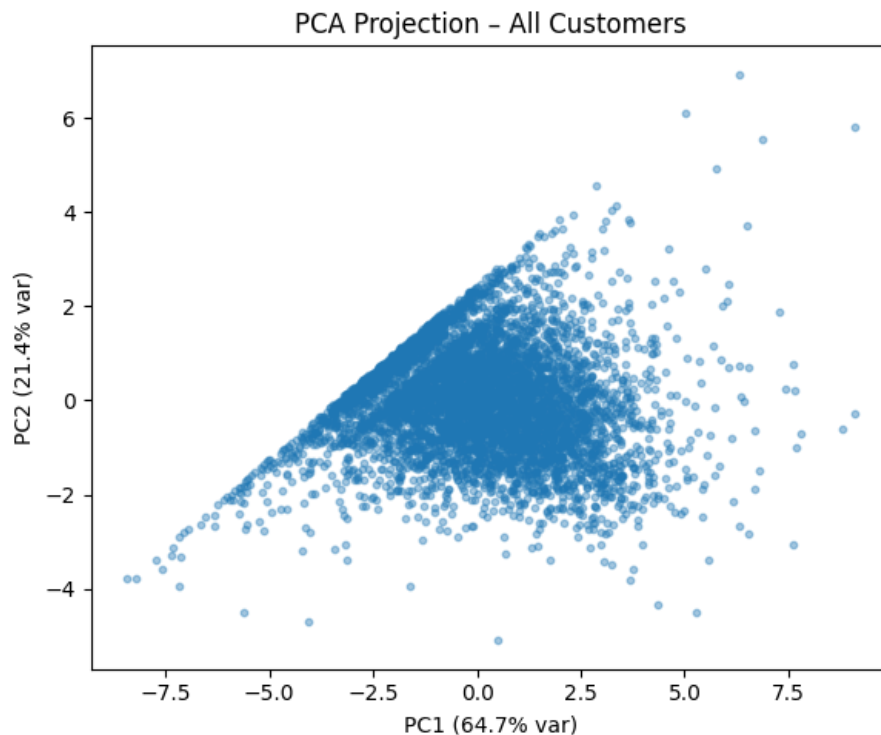


Figure 1: PCA Projection — All Customers. PC1 (64.7% variance) separates customers by order volume and total spend; PC2 (21.4% variance) captures recency and frequency variation. The fan-shaped distribution suggests two distinct behavioural groups.

2.3 KMeans Tuning

The optimal number of clusters k was selected by computing the silhouette score for $k = 2$ through 8. The full results are shown in Figure 2 and tabulated below:

k	Silhouette Score
2	0.321
3	0.254
4	0.232
5	0.225
6	0.217
7	0.206
8	0.198

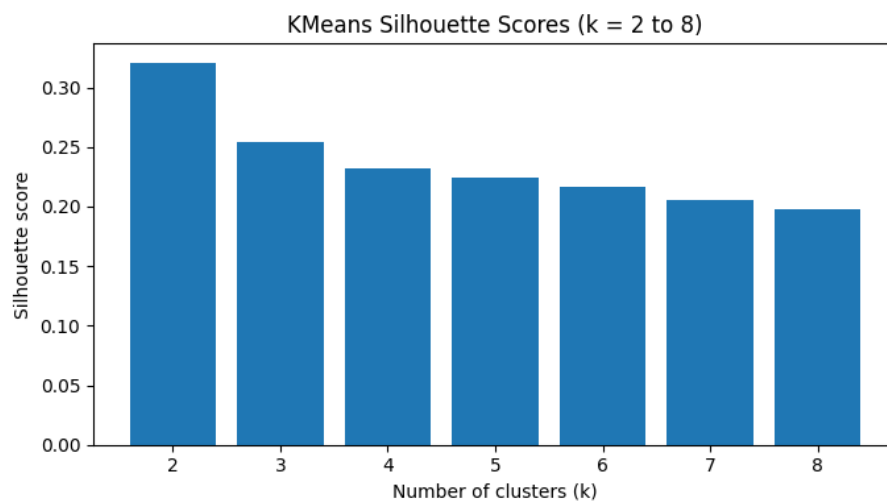


Figure 2: KMeans Silhouette Scores for $k = 2$ to 8. The score is strictly decreasing after $k = 2$, and the largest single drop is from $k = 2$ to $k = 3$, confirming that two clusters represent the natural partition of this dataset.

The silhouette score is strictly decreasing after $k = 2$, and the drop from $k = 2$ to $k = 3$ is the largest single step (0.321 to 0.254). This clearly indicates that $k = 2$ is the natural partition in this data — adding more clusters fragments existing groups rather than revealing new structure. Best K was therefore set to 2.

2.4 Clustering Algorithm Comparison

KMeans with $k = 2$ achieved a silhouette score of 0.321. Hierarchical Agglomerative Clustering (HAC) with $k = 2$ and average linkage achieved a silhouette score of 0.604. The code programmatically compares both scores and selects the better model — **HAC was selected**.

Average linkage was chosen because it computes inter-cluster distance as the average of all pairwise distances between two clusters, making it more robust to outliers than complete linkage and less prone to chaining than single linkage. This is appropriate for a dataset where extreme wholesale customers exist alongside a dense mass of typical retail buyers.

The performance gap (0.321 vs 0.604) is substantial and expected. KMeans assumes spherical, equally-sized clusters — both assumptions are violated by this dataset. The fan-shaped PCA distribution is non-spherical, and the extreme wholesale buyers distort KMeans cluster centres. HAC’s average linkage handles the elongated structure and extreme points more gracefully.

DBSCAN was also tested. Using the 5-nearest-neighbour distance plot to guide eps selection, the algorithm was run with `eps = 0.268` and `min_samples = 5`. The result was 52 clusters and 4,882 noise points, meaning approximately 83% of customers were classified as noise. This confirms that DBSCAN is unsuitable for this dataset. The customer feature space does not exhibit the kind of uniform-density, clearly separated regions that DBSCAN requires. DBSCAN was therefore excluded from the final analysis.

2.5 Cluster Naming and Interpretation

Cluster naming was performed programmatically by comparing median Monetary across clusters. The cluster with the higher median Monetary was assigned the name **Wholesale Buyers** and the remaining cluster was named **Regular Buyers**. The cluster profiles (median values) are:

Feature	Regular Buyers	Wholesale Buyers
Customers	5,871	7
Monetary (£)	896.60	12,393.70
Frequency	3	2
AvgOrderValue (£)	284.87	10,877.18
AvgQtyPerInvoice	157.5	17,766.0
Variety	45	9

Regular Buyers are diverse, moderately frequent, and relatively low-value per visit. Wholesale Buyers are rare, generate 14 times more revenue per customer, place orders of over 17,000 units per invoice on average, and concentrate their purchasing on just 9 product types. These two groups require entirely different commercial strategies: loyalty programmes, cross-sell campaigns, and personalised recommendations for Regular Buyers; dedicated account management, volume pricing, and supply chain planning for Wholesale Buyers.

3 Segment-Specific Market Basket Analysis

3.1 Approach

The original transactional data was split into subsets based on cluster assignments. The Apriori algorithm was run separately on each subset to discover association rules, ranked by lift. Lift greater than 1 indicates that two items are bought together more often than expected by chance; the higher the value, the more actionable the rule. Clusters with fewer than 30 customers or fewer than 50 invoices were automatically skipped.

3.2 Results

Regular Buyers (Cluster 0): The basket matrix comprised 31,212 invoices across 165 items. The top five rules by lift are summarised in Table 2.

Table 2: Top Association Rules — Regular Buyers

Antecedent	Consequent	Supp.	Conf.	Lift
PINK REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAUCER	0.020	0.838	28.03
GREEN REGENCY TEACUP AND SAUCER	PINK REGENCY TEACUP AND SAUCER	0.020	0.670	28.03
ROSES REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAUCER	0.024	0.704	23.54
GREEN REGENCY TEACUP AND SAUCER	ROSES REGENCY TEACUP AND SAUCER	0.024	0.796	23.54
DOLLY GIRL LUNCH BOX	SPACEBOY LUNCH BOX	0.022	0.673	18.97

A lift of 28.03 on the Regency Teacup pairing means a customer who buys the green teacup is 28 times more likely to also buy the pink one than a random customer would be. A confidence of 0.838 means that in 84% of cases where a customer bought the pink teacup, they also bought the green one. This is textbook set-completion behaviour: these items form a matching decorative collection and customers are deliberately assembling the full set. The retailer should bundle these items on the product page, display them in a “Complete the Collection” section, and trigger recommendation emails to any customer who purchases one piece but not the others.

The Lunch Box rule (DOLLY GIRL $\rightarrow$ SPACEBOY, lift = 18.97) reflects the same logic across children’s novelty gifts, likely driven by parents or gift-buyers purchasing for multiple children.

Wholesale Buyers (Cluster 1): Only 7 customers and 16 invoices — correctly flagged as too small for Apriori and skipped. Wholesale purchasing logic is driven by volume and margin rather than product affinity, so the absence of basket rules for this segment is the correct finding.

4 Time-Series Decomposition

4.1 Cluster and Item Selection

The code programmatically selects the cluster meeting minimum size thresholds (30 or more customers, 50 or more invoices) ranked by median Monetary. Regular Buyers was selected: 5,871 customers and 36,953 invoices. Within this cluster, the item with the highest total quantity sold and sufficient purchase history (at least 14 distinct purchase days and at least 30 unique invoices) was identified programmatically as **WORLD WAR 2 GLIDERS ASSTD DESIGNS**.

4.2 Decomposition

Sales of this item were aggregated by day and decomposed using STL (Seasonal and Trend decomposition using Loess) with `period=7` and `robust=True`. The 99th-percentile clipping applied before decomposition prevents a small number of extremely large individual orders from distorting the seasonal and trend estimates.

The **trend** component shows medium-term fluctuations rather than a monotonically increasing trajectory, indicating a genuine product popularity cycle driven by promotional activity or seasonal restocking behaviour. The non-monotonic trend means that inventory planning based solely on year-over-year averages would be misleading.

The **seasonal** component shows a clear, consistent weekly pattern throughout the entire series. The regularity of the weekly peaks and troughs confirms that day-of-week is a genuine and reliable driver of purchasing behaviour, consistent with a product ordered on specific days for mid-week delivery or bought as a weekend impulse gift.

The **residual** component is mean-zero but contains large occasional spikes corresponding to events outside the weekly seasonal model, likely promotional campaigns, school holidays, or unusually large one-off orders. The magnitude of these spikes relative to the seasonal amplitude indicates that event-driven demand is a significant contributor to total sales.

4.3 Business Insight

Inventory replenishment schedules should be aligned to the weekly demand peak to minimise stockout risk, and order quantities should track the current position in the medium-term trend cycle. The large residual spikes support the value of targeted event-driven promotional campaigns: pre-planned activity around school holidays and seasonal gifting periods could capture these demand surges reliably.

5 Anomaly Detection

5.1 Method

Local Outlier Factor (LOF) was applied to the scaled customer feature table with `n_neighbors = 20` and `contamination = 0.01`. LOF is a density-based algorithm that compares the local density of each data point to the density of its k nearest neighbours. A customer whose neighbourhood is much less dense than their neighbours' receives a high LOF score and is flagged as an outlier. Unlike simple threshold-based methods, LOF detects customers who are locally unusual relative to behaviourally similar customers. With 5,878 customers and `contamination = 0.01`, the algorithm flagged **59 anomalies**: 52 from Regular Buyers and 7 from Wholesale Buyers.

5.2 Top 10 Anomalous Customers

Table 3 summarises the top 10 most anomalous customers by LOF score.

Table 3: Top 10 Anomalous Customers

Customer ID	LOF Score	Cluster	Key Flags
17961.0	2.976	Regular Buyers	high frequency
16446.0	2.607	Wholesale Buyers	high monetary, low variety, high avg order
15098.0	2.587	Regular Buyers	high monetary, low variety, high avg order
17850.0	2.406	Regular Buyers	high frequency, high monetary
13734.0	2.327	Regular Buyers	multivariate anomaly
13687.0	2.302	Wholesale Buyers	high avg order, bulk buying, huge single line
18037.0	2.185	Regular Buyers	multivariate anomaly
12748.0	2.144	Regular Buyers	high frequency, high monetary, huge single line
12918.0	2.053	Regular Buyers	low variety, high avg order
17988.0	2.022	Regular Buyers	multivariate anomaly

5.3 Detailed Interpretation

Customer 17961.0 (Regular Buyers, LOF = 2.976) placed 100 unique invoices — the most distinguishing feature — generating £2,995.54 across 486 unique products. Within the Regular Buyers cluster where median frequency is 3 invoices, 100 invoices is structurally isolated from all nearest neighbours. The pattern of very high frequency combined with very low spend per visit suggests either a staff account, an automated test account, or a customer systematically placing many small sampling orders.

Customer 16446.0 (Wholesale Buyers, LOF = 2.607) placed only 2 invoices but generated £168,472.50 in total revenue. The raw transaction data shows a single line item of

80,995 units generating £168,469.60 in one transaction, with Variety of just 3 products. Even within the Wholesale Buyers cluster, this profile is extreme and warrants immediate investigation as either a major distributor's one-time bulk drop-shipment or a transaction requiring fraud and data integrity review.

Customer 15098.0 (Regular Buyers, LOF = 2.587) placed 3 invoices generating £39,916.50 with an AvgOrderValue of £13,305.50 and Variety of 1 unique product. A customer spending nearly £40,000 on a single product category across 3 visits is highly anomalous within the Regular Buyers cluster where typical customers spend £285 per visit across 45 product types.

Customer 17850.0 (Regular Buyers, LOF = 2.406) placed 155 invoices generating £56,600.08 across 100 unique products. Both frequency and Monetary sit at the extreme top end of the Regular Buyers distribution, and this customer may represent a small reseller incorrectly classified in the retail segment.

Customer 13734.0 (Regular Buyers, LOF = 2.327) presents a multivariate anomaly: no single feature is individually extreme by the flagging thresholds, but the combination of an AvgOrderValue of £1,675.41, modest Variety, and infrequent but high-value visits places this customer in a low-density region of the feature space that single-variable threshold rules would miss entirely.

Customer 13687.0 (Wholesale Buyers, LOF = 2.302) placed 1 invoice with a maximum single-line quantity of 7,128 units and a total of 87,167 units — the highest AvgQuantityPerInvoice in the dataset. A single invoice for 87,167 units is extreme even within the Wholesale Buyers cluster and likely represents a distributor drop-shipment or a bulk contract order processed as a single transaction.

Customer 18037.0 (Regular Buyers, LOF = 2.185) placed 15 invoices generating only £167.42 in total, an AvgOrderValue of £11.16, purchasing 31 unique products with a maximum line quantity of just 2 units. Moderate frequency combined with extremely low spend per visit and very small quantities per line is a multivariate profile far from any typical Regular Buyer neighbour.

Customer 12748.0 (Regular Buyers, LOF = 2.144) placed 336 invoices — the highest frequency in the dataset — generating £56,599.39 across 2,286 unique products and 39,108 total units, with a maximum single-line quantity of 576. Active across the full two-year dataset window, this profile most plausibly represents a business or internal purchasing account placing frequent operational orders.

Customer 12918.0 (Regular Buyers, LOF = 2.053) placed 1 invoice for 1 unit generating £10,953.50 — a unit price of £10,953.50. With Variety of 1 product and Recency of 626 days, this is almost certainly a large one-off corporate gift order, a bulk pricing contract recorded as a single-unit transaction, or a data entry error in the Price field.

Customer 17988.0 (Regular Buyers, LOF = 2.022) placed 14 invoices generating £489.54

across 115 unique products. Buying from 115 different product categories while spending only £35 per invoice is unusual: typical neighbours either spend more for that level of variety or have far lower variety for that spend level.

5.4 Cluster Distribution and Business Priorities

Of the 59 flagged anomalies, 52 are Regular Buyers and 7 are Wholesale Buyers. Notably, all 7 members of the Wholesale Buyers cluster were flagged, confirming that each wholesale customer has a highly distinctive individual profile even compared to other wholesale customers. The sales team should prioritise reviewing Customers 16446.0, 13687.0, 15098.0, and 12918.0 as the highest-priority cases for data quality verification and potential fraud review. Customers 17850.0 and 12748.0 may warrant outreach as potentially misclassified trade buyers who could be better served under a wholesale account structure.

6 Recommendation System (SVD)

6.1 Data Preparation

To build a recommendation system, the transactional data was transformed into an implicit ratings matrix. Each (Customer ID, StockCode) pair was aggregated by summing total quantity purchased, then clipped to the range $[1, 5]$ to simulate a rating scale — a customer buying more units of an item is treated as giving it a higher implicit rating. Because the full ratings table exceeds 50,000 rows, a 30% stratified random sample of users was drawn using a fixed random seed of 42 for hyperparameter tuning, with the full dataset used to train the final model.

6.2 Hyperparameter Tuning and Baseline Comparison

The `n_factors` parameter was tuned over $\{1, 2, 3, 4, 5\}$ using 3-fold cross-validation with RMSE as the evaluation metric. The full tuning results are summarised in Table 4.

Table 4: SVD Model Performance by `n_factors`

<code>n_factors</code>	CV RMSE	Rank
1	1.1041	1
2	1.1046	2
3	1.1052	4
4	1.1051	3
5	1.1057	5

The optimal model uses `n_factors` = 1 with a cross-validated RMSE of 1.1041. To confirm that the SVD model adds genuine value beyond a naive approach, a `BaselineOnly` model — which predicts ratings using only global mean, user bias, and item bias with

no latent factors — was evaluated under the same cross-validation setup. The baseline achieved $\text{RMSE} = 1.1230$. The SVD model outperforms the baseline by 0.0189, a relative improvement of approximately 1.7%, confirming that the latent factor structure captures real signal beyond simple popularity and user-level biases.

6.3 Interpretation

The fact that `n_factors = 1` is optimal is itself a meaningful finding. A single latent dimension captures customer-item interactions better than any higher-rank decomposition, meaning that adding more factors introduces more noise than signal. The tight clustering of RMSE values across all five options (range of only 0.0016) confirms that the data's preference structure is genuinely low-dimensional. This aligns directly with the clustering results: the customer base splits into two behaviourally distinct groups, and within those groups purchasing patterns are relatively homogeneous.

6.4 Business Application

The final SVD model was trained on the full ratings dataset and is ready for deployment. In production it can generate a ranked list of recommended items for any `Customer ID`, surfacing products the customer has not yet purchased but that similar customers buy. Training separate SVD models per cluster — recommending bulk-order SKUs to Wholesale Buyers and decorative gift sets to Regular Buyers — represents the most valuable next iteration.

Final Business Summary

Executive Summary

The analysis reveals a clear two-segment customer structure: a large group of **Regular Buyers** (5,871 customers) and a small but high-impact group of **Wholesale Buyers** (7 customers). Regular Buyers exhibit diverse purchasing behaviour with moderate spending, while Wholesale Buyers generate significantly higher revenue through a small number of extremely large orders.

Market basket analysis identified strong product associations within Regular Buyers, particularly in decorative product sets such as Regency teacups, with lift values reaching 28.03, indicating highly reliable cross-sell opportunities.

Time-series analysis of the top-selling item showed clear weekly seasonality, confirming that demand consistently peaks on specific days of the week and should guide both inventory planning and promotional scheduling.

Anomaly detection identified 59 unusual customers, including all 7 Wholesale Buyers, highlighting cases of extreme purchasing behaviour that require business review.

The SVD recommendation model achieved a 1.7% RMSE improvement over the naive baseline, demonstrating meaningful predictive capability while confirming that the retailer's implicit preference structure is low-dimensional.

Business Interpretation

This project demonstrates how unsupervised learning techniques can be combined into a coherent pipeline that delivers actionable insights at each stage.

- **Customer segmentation** enables targeted commercial strategies tailored to each behavioural group.
- **Association rules** support high-confidence bundling and cross-selling, particularly “Complete the Collection” campaigns for decorative product sets.
- **Time-series analysis** improves inventory planning by quantifying weekly seasonality and medium-term demand cycles.
- **Anomaly detection** enhances risk control by surfacing customers with extreme or unusual purchasing patterns for review.
- **Recommendation modelling** enables personalisation at scale, with segment-specific models as the recommended next step.

Together, these results illustrate the value of transforming raw transactional data into structured, evidence-based strategic business intelligence.